

Assessment

N+ - Network security, Maintenance and Troubleshooting procedures

Section A — Concept Application

Question: What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

ANS :-

1. CPU (The Brain) :-

- Performance : Match cores and speed to your work (4-6 cores for basic tasks, 8+ for heavy gaming or editing).
- Compatibility : Must match your motherboard's socket type (e.g., Intel LGA1700 or AMD AM5).

2. RAM (Short-Term Memory) :-

- Performance: Get at least 16 GB for smooth multitasking.
- Compatibility: Must match motherboard generation (DDR4 vs. DDR5) and CPU speed limits.

3. Storage (Long-Term Memory) :-

- Performance: Choose an NVMe SSD over slow hard drives for fast startup and loading.
- Compatibility: Check if your motherboard has an open M.2 slot and supports the SSD generation (PCIe 3.0, 4.0, or 5.0).

Question : Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

ANS :-

CPU (Brain) :

Constantly analyzes background processes, scans new code, and inspects files in real time, leaving fewer processor cycles for your apps.

RAM (Memory) :

Stays loaded in the background with threat databases, taking up active memory that your browser or games need.

Storage (Drive) :

Reads every single file you open, save, or download before letting you use it, causing high drive usage and slow load times.

Question : Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

ANS :-

1.Boot into Safe Mode :

Restart your computer in Safe Mode to load only essential files and isolate the problem.

2.Update or Roll Back Drivers :

Open Device Manager :

- If the issue started after an update, click Roll Back Driver.
- If a driver is outdated or missing, click Update Driver or reinstall it directly from the manufacturer's site.

3.Run OS Repair Tools :

Open Command Prompt as administrator and run `sfc /scannow` to scan and repair corrupted operating system files.

4.Use System Restore :

Revert your operating system to a System Restore Point from a date before the issue started.

Question : Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

ANS :-

Surge Protector (Shield) :

Blocks sudden voltage spikes (like lightning) so your internal hardware doesn't fry.

UPS (Emergency Battery) :

Keeps your PC running during power cuts so you have time to save files and shut down safely without corrupting data.

Question : Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

ANS :-

1. How They Work Together :-

- DHCP (Address Giver) : Hands your PC a network identity (IP address) so it can connect to the router.
- DNS (Phonebook) : Translates website names (like google.com) into numerical IP addresses so your browser can load them.

2. Which One Is Misconfigured?

- It's DNS if : You can't open websites using name URLs (like google.com), but you can load them using their direct IP address (like 8.8.8.8).
- It's DHCP if : Your device shows "No Internet," "Self-assigned IP" (169.254.x.x), or cannot connect to the local network at all.

Question : Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

ANS :-

1. How to Determine (Troubleshoot) :-

- Check Firewall Logs : Look for "DROP" or "DENY" messages matching your app's IP address or port.
- Test-Disable Briefly : Turn the firewall off for 1 minute. If traffic works, the firewall rules are definitely the problem.
- Trace Port Traffic : Run telnet [IP] [Port] or use Test-NetConnection to see if the connection times out.

2. How to Resolve :-

- Add an Allow Rule : Create an Inbound/Outbound rule permitting that specific IP address, application, or port.
- Reorder Rules : Move your new "Allow" rule above any "Deny All" rules (firewalls check rules top-to-bottom).

Section B — Practical Application

Task 1: Workstation Assembly

ANS :-

1.Prepare the Motherboard (Out of Case) :

- CPU : Align gold triangle, drop CPU into socket, and latch lever.

- Cooler : Apply thermal paste (if not pre-applied) and mount CPU cooler.
- RAM : Push sticks into slots until clips snap locked.
- M.2 SSD : Insert into slot at an angle, press flat, and secure with screw.

2.Mount Power Supply & Motherboard :

- Screw the PSU into the bottom back of the chassis.
- Install motherboard standoffs inside the case, align the I/O shield, and screw down the motherboard.

3.Connect Cables :

- Power : Plug in 24-pin main power, CPU power (8-pin), and GPU power.
- Front Panel : Connect Power Switch, Reset, Audio, and USB headers to motherboard pins.
- Storage/Fans : Connect SATA data cables (for 2.5"/3.5" drives) and fan headers.

4.Install GPU (Graphics Card) :

- Slot the GPU into the top PCIe slot until it clicks, then screw it into the rear PC expansion bracket.

5.Power ON & Test POST :

- Plug in monitor to GPU, connect power cable, flip PSU switch ON, and press power button.
- Success (POST Passed) : PC beeps/lights turn green and displays BIOS screen.

Task 2: RAM Mismatch Troubleshooting

ANS :-

1.Identify Error at POST :

Watch the screen for POST warnings like "Memory Size Decreased" or "Memory Configuration Error".

2.Determine Cause :

Check BIOS specs vs physical stick labels :

- Speed/Timing : Sticks running at different MHz or latencies.
- Capacity/Slots : Sticks installed in wrong channels (e.g., slot 1 & 3 instead of 2 & 4).

3.Fix Hardware & Settings :

- Reseat : Turn off PC, remove sticks, and click them back into correct dual-channel slots.
- Auto-Match : In BIOS, enable XMP/EXPO or set RAM speed/timings to match the slowest stick.

4.Confirm Normal Boot :

Save BIOS settings and restart. Check Task Manager to confirm full RAM capacity and speed are recognized.

Task 3: LAN Topology Build

ANS :-

1.Build Topology :

- Drag 1 Switch to the workspace.
- Drag 5 PCs and 1 Printer.

2.Connect Cables :

- Use Copper Straight-Through cables to connect all 5 PCs and the Printer to the Switch ports.

3.Assign Class C Ips :

Click each device → Desktop / Config tab → IP Configuration :

- PC 1: 192.168.1.1
- PC 2: 192.168.1.2
- PC 3: 192.168.1.3
- PC 4: 192.168.1.4
- PC 5: 192.168.1.5
- Printer: 192.168.1.6
- *Subnet Mask for all: 255.255.255.0*

4.Test Connectivity (Ping) :

- Click PC 1 → open Command Prompt.
- Type ping 192.168.1.6 to ping the Printer .
- Success: You will see *"Reply from 192.168.1.6: bytes=32 time<1ms TTL=128"*.

Task 4: Clean OS Installation

ANS :-

1.Boot from USB :

- Create a bootable USB from the ISO .
- Plug in USB, restart PC, and press F12 / F2 / Del to select USB in Boot Menu.

2.Manual Partitioning :

Select Custom Install (Manual Partitioning) :

- Windows : Delete old partitions → click New to create a main drive partition for OS → select it and click Next.
- Linux : Create 3 partitions manually : /boot/efi, swap, and / (Root, remaining space, ext4).

3.Complete Setup & Boot :

- Let setup install files, reboot, remove USB, and set up your user account until you reach the desktop error-free.

4.Optimize Startup :

- Windows : Open Task Manager (Ctrl + Shift + Esc) → Startup apps tab → Disable heavy non-essential apps.
- Linux : Open Startup Applications or run `systemctl disable [service]` to stop unneeded background services.

ANS :-

1. Top 3 Components Causing Reboots :-

- Power Supply (PSU) :
Fails to deliver stable power when GPU/CPU demand spikes.
- CPU Cooling System :
Thermal shutoff triggers to prevent CPU heat damage.
- RAM Modules :
Faulty memory causes instant crashes when under heavy load.

2. Why POST Passes in Morning but Fails in Afternoon :-

- Thermal Accumulation :
Cold hardware passes POST after standing overnight. By afternoon, high ambient room temperatures and built-up case heat cause the system to instantly overheat under load.

3. Stress-Test Confirmation :-

- Run a stress tool (like OCCT or Prime95 + FurMark) :
 - Thermal Issue :
Temps climb rapidly over 90°C–95°C before system shuts down.
 - Power Issue :
Instant black screen reboot as soon as GPU + CPU hit 100% load, with normal temps.

4. 3 Remediation Steps :-

1. Repaste & Clean :
Reapply CPU thermal paste and clean dust from heatsinks/fans.

2. Improve Airflow :

Add case fans and fix cable management for better ventilation.

3. Reset RAM Settings :

Disable XMP/EXPO or run MemTest86 to isolate bad memory sticks.

5. Is a PSU Upgrade Necessary? :-

- Yes, IF :

Total system wattage exceeds your PSU rating, or your current PSU uses degraded components that trigger Over-Current Protection (OCP) under power spikes.

- No, IF :

Reboots stop after fixing temperatures or thermal throttling.

ANS :-

***Prompt Given to AI :-**

"Create a simple IP addressing and subnetting plan for Nexus Innovations' startup office with 5 systems and 1 printer. Include subnet, subnet mask, usable host range, broadcast address, IP address for each device, and addresses reserved for future expansion."

*** AI Original Plan vs. Corrected Version :-**

Device	AI's Original Plan	Corrected Plan
Subnet ID	192.168.1.0/29	192.168.1.0/28
Subnet Mask	255.255.255.248	255.255.255.240
Usable Host Range	192.168.1.1 to 192.168.1.6 (6 hosts)	192.168.1.1 to 192.168.1.14 (14 hosts)
Broadcast Address	192.168.1.7	192.168.1.15
Gateway Router	192.168.1.1	192.168.1.1
2PC 1 to PC 5	192.168.1.2 – 192.168.1.6	192.168.1.2 – 192.168.1.6
Printer	192.168.1.7 <i>(Invalid! Same as broadcast)</i>	192.168.1.7
Future Expansion	None available (0 addresses left)	192.168.1.8 – 192.168.1.14 (7 reserved IPs)

***Correction Note :-**

- **What changed :** Upgraded the subnet from /29 to /28 and fixed the printer's IP address.
- **Why it was needed :** The AI's /29 subnet only provided 6 usable host IPs (1 for router, 5 for PCs), leaving no IP for the printer (assigned to the broadcast address 192.168.1.7) and zero space for growth. Changing to /28 provides 14 usable IPs, fixing the IP conflict and reserving 7 IPs for future expansion.